

Optimization: Critical Points & Lagrange Multipliers

ANSWER KEY



Finding critical points

- 1 Find the critical point(s) of $f(x, y) = x^2 + 3y^2 - 4x + 6y$.
 $f_x = 2x - 4 = 0 \Rightarrow x = 2$. $f_y = 6y + 6 = 0 \Rightarrow y = -1$. Critical point: $(2, -1)$.
- 2 Find the critical point(s) of $f(x, y) = xy - x^2 - y^2$.
 $f_x = y - 2x = 0 \Rightarrow y = 2x$. $f_y = x - 2y = 0 \Rightarrow x = 2y$. Substituting: $x = 2(2x) = 4x \Rightarrow x = 0$, so $y = 0$.
Critical point: $(0, 0)$.

The Hessian and the Second Derivative Test

- 3 Classify the critical point $(2, -1)$ of $f(x, y) = x^2 + 3y^2 - 4x + 6y$ using $D = f_{xx}f_{yy} - (f_{xy})^2$.
 $f_{xx} = 2$, $f_{yy} = 6$, $f_{xy} = 0$. $D = 2(6) - 0 = 12 > 0$ and $f_{xx} > 0$, so $(2, -1)$ is a local minimum.
- 4 Classify the critical point $(0, 0)$ of $f(x, y) = xy - x^2 - y^2$.
 $f_{xx} = -2$, $f_{yy} = -2$, $f_{xy} = 1$. $D = (-2)(-2) - 1^2 = 3 > 0$ and $f_{xx} < 0$, so $(0, 0)$ is a local maximum.
- 5 Classify the critical point of $f(x, y) = x^2 - y^2$.
 $f_x = 2x = 0$, $f_y = -2y = 0 \Rightarrow (0, 0)$. $f_{xx} = 2$, $f_{yy} = -2$, $f_{xy} = 0$. $D = 2(-2) - 0 = -4 < 0$, so $(0, 0)$ is a saddle point.

Lagrange multipliers: setting up the system

- 6 Set up the Lagrange multiplier equations to optimize $f(x, y) = xy$ subject to $g(x, y) = x + y - 10 = 0$.
 $\nabla f = \lambda \nabla g$: $\langle y, x \rangle = \lambda \langle 1, 1 \rangle \Rightarrow y = \lambda$, $x = \lambda \Rightarrow x = y$, together with the constraint $x + y = 10$.

Solving Lagrange multiplier problems

- 7 Maximize $f(x, y) = xy$ subject to $x + y = 10$.
From the setup above, $x = y$ and $x + y = 10 \Rightarrow x = y = 5$. Maximum value: $f(5, 5) = 25$.
- 8 Minimize $f(x, y) = x^2 + y^2$ subject to $x + y = 4$.
 $\nabla f = \lambda \nabla g$: $\langle 2x, 2y \rangle = \lambda \langle 1, 1 \rangle \Rightarrow x = y$. With $x + y = 4$: $x = y = 2$. Minimum value: $f(2, 2) = 8$.

A geometric Lagrange multiplier application

- 9 Find the point on the line $x + y = 6$ closest to the origin, using Lagrange multipliers on $f(x, y) = x^2 + y^2$.
 $\nabla f = \lambda \nabla g: (2x, 2y) = \lambda(1, 1) \Rightarrow x = y$. With $x + y = 6$: $x = y = 3$. Closest point: $(3, 3)$, at distance $\sqrt{9 + 9} = 3\sqrt{2}$.